

# INDEX

# Analog Circuit

NAME: Sohail Ansari STD.: \_\_\_\_\_ SEC.: \_\_\_\_\_ ROLL NO.: \_\_\_\_\_ SUB.: Maths

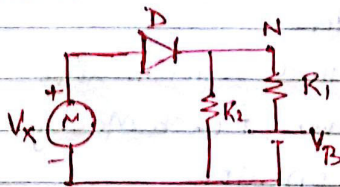
[illegible]

# LECTURE : 09

classmate

Date  
Page 27

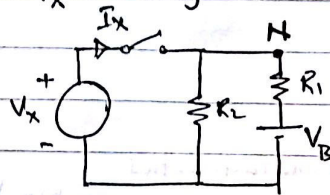
Example :



Plot ①  $I_x$  Vs  $V_x$

②  $V_N$  Vs  $V_x$

When  $V_x$  is very -ve - then  $D \rightarrow$  off



$I_x = 0$

When Diode is in state of getting turn on Current  $I_x$  is very less so, we assume that current is not going to  $R_2$  &  $R_1$  branches. When  $V_x$  is  $V_{x1}$  then Diode begins to turn on but current is very small.  $\therefore V_{x1} = V_N + 0.7 = \cancel{V_B + 0.7}$

where

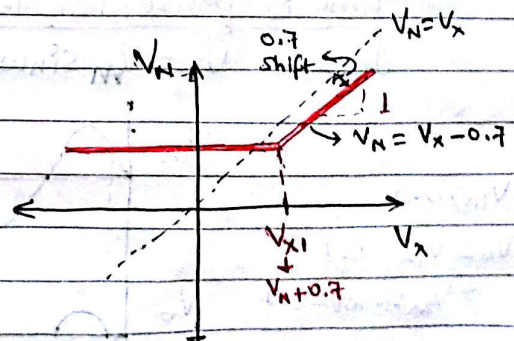
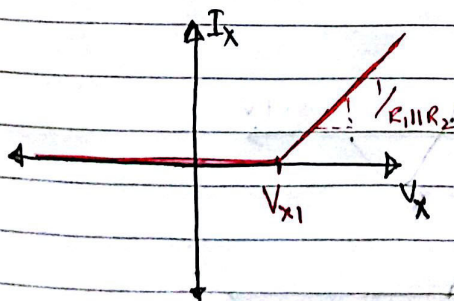
$$V_N = \cancel{-\frac{V_B R_1}{R_1 + R_2}} + V_B = V_B \left[ 1 + \frac{R_1}{R_1 + R_2} \right]$$

$\therefore$  When  $V_x > (V_N + 0.7)$

$$\cancel{I_x = V_x - 0.7 - V_N} = \cancel{V_x - 0.7 - V_B \left[ 1 + \frac{R_1}{R_1 + R_2} \right]}$$

Use Nodal to find  $I_x$

$$I_x = \left( \frac{1}{R_2} + \frac{1}{R_1} \right) (V_x - 0.7) - \frac{V_B}{R_1}$$



When  $V_x < V_N + 0.7$   $D \rightarrow$  off

$$\cancel{V_N = V_B \left[ 1 + \frac{R_1}{R_1 + R_2} \right]}$$

$$V_N = \frac{R_1 V_B}{R_1 + R_2}$$

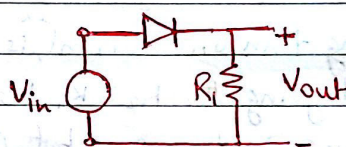


## PRACTICAL DIODE CIRCUITS

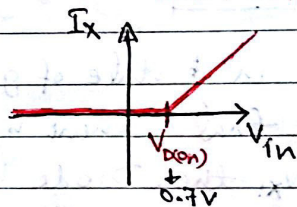
1. Rectifiers  $\rightarrow$  charger, Adapters
2. Limiters  $\rightarrow$  FM receivers, optical communication.
3. Voltage Doublers  $\rightarrow$  RF receivers, Flash Memory.
4. Level shifts  $\rightarrow$  General Circuits.

### 1 RECTIFIERS

#### (a) Half Wave Rectifier



I-V characteristics



When  $V_{in} < 0.7$

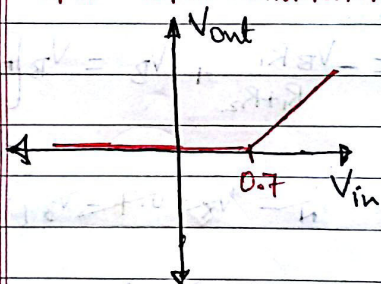
$D \rightarrow \text{OFF}$

$I = 0$

When  $V_{in} > 0.7$

$$I_x = \frac{V_{in} - 0.7}{R_L}$$

Input-output characteristics



When  $V_{in} < 0.7V \rightarrow D \rightarrow \text{O.C.}$

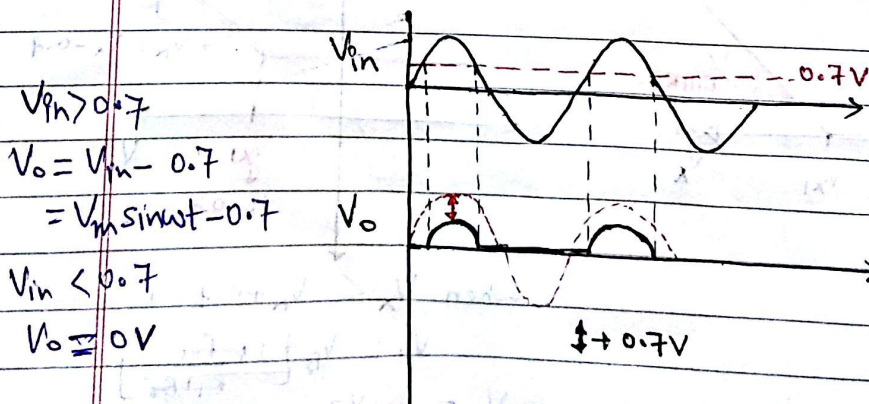
$I = 0, V_{out} = IR = 0$

When  $V_{in} > 0.7V \rightarrow \text{Diode} \rightarrow \text{S.C.}$

$$V_{out} = V_{in} - 0.7$$

#### Time Response characteristics

In this  $V_{in} = V_m \sin \omega t$



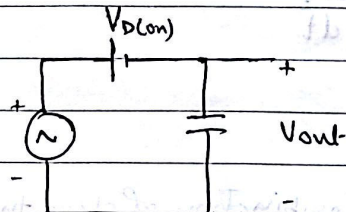
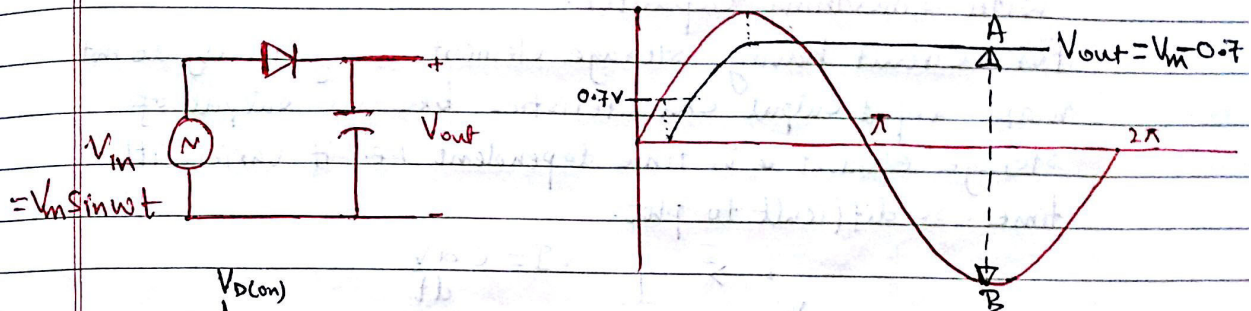
# LECTURE : 10

classmate

Date

Page 29

## Half wave Rectifier with Smoothing Capacitor

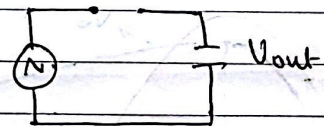


When  $V_{in} > V_{D(on)}$  Diode turns on  
Capacitor charges up to Max value  
When  $V_{in}$  again goes after its  
Max. value then voltage across capacitor  
is  $V_m - V_{D(on)} = V_m - 0.7V$   
 $\therefore V_{out} > V_{in}$  in this case & diode  
turns off

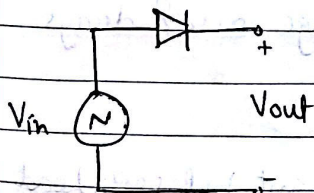
### Reverse voltage across Diode:

When Diode is Reverse biased

$$V_{BR} = V_{BR} = AB = 2V_m - 0.7$$



### Role of $R_L$ in original circuit:

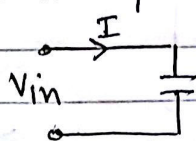


It is incorrect to say  
that  $V_{out} = 0$  when  
Diode is open. A voltage  
can be zero only through finite impedance  
not infinite impedance. So it is  
necessary to attach any element first to  
make sure  $V_{out} = 0$  when Diode is off.



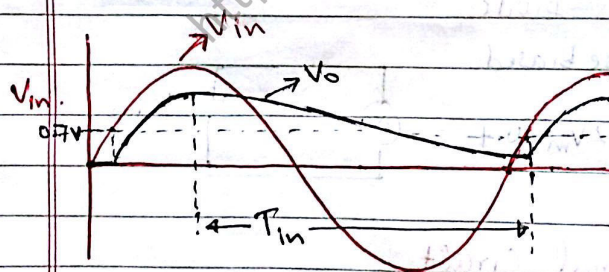
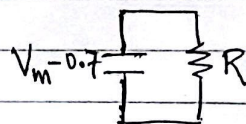
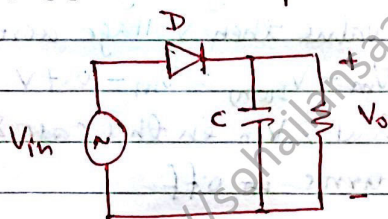
- Can we plot input-output characteristics for rectifier with smoothing capacitor.

For circuit having storage element, we generally do not make input output characteristics because output of storage element is time dependent i.e. it varies with time. So difficult to plot.



$$I = C \frac{dV}{dt}$$

- HWR with Capacitor & Load [Combination of above two] Case



After peak diode turns off. Then capacitor gives energy to Load. It decays exponentially unless & until diode turns on again.

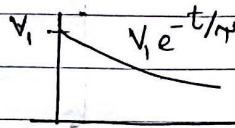
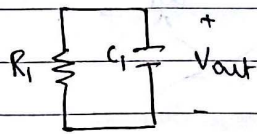
Because of resistance output voltage rise & decays. Means output voltage has ripple.

Conclusion:

→ without Load output voltage is constant but with Load it draws current & output voltage has some ripple. Because of this ripple humming noise from output we get. Ex: In speaker.

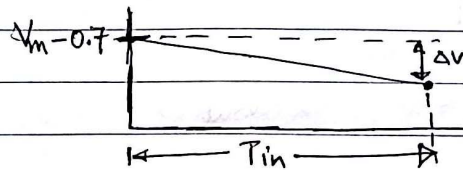
→ If ripple is small then there would be less decay during  $T_{in}$ .

• Ripple calculation



If  $t \ll \tau$

then we can approximate exponential to straight line.



$$V_{out} = (V_m - 0.7) \left(1 - \frac{t}{\tau}\right)$$

Decay behaviour.

$$\therefore V_{out}(at t = T_{in}) =$$

$$= (V_m - 0.7) \left(1 - \frac{T_{in}}{\tau}\right)$$

$$\text{Now, Ripple Amplitude } \Delta V = (V_m - 0.7) - \left[(V_m - 0.7) \left(1 - \frac{T_{in}}{\tau}\right)\right]$$

$$\Delta V = (V_m - 0.7) \frac{T_{in}}{\tau}$$

$$T_{in} \approx T = \frac{1}{f_{in}}$$

$$\Delta V = \frac{(V_m - V_{D(on)})}{R_1 C_1 f_{in}}$$

$C \uparrow, \Delta V \downarrow$

$R \downarrow$  (means heavier load)

then  $\Delta V \uparrow$ .

From GATE point of view taking ideal diode

HALF WAVE RECTIFIER

$$V_o = \frac{1}{2\pi} \int_0^\pi V_m \sin \omega t = V_m / \pi$$

$$V_{orms} = \sqrt{\frac{1}{2\pi} \int_0^\pi V_m^2 \sin^2 \omega t} = V_m / 2$$

$$V_{BR} = 2V_m = \text{Peak inverse voltage.}$$

$$\Delta V = \frac{V_m}{RCf} \quad (\text{taking ideal diode})$$

$$f_{out} = f_{in}$$

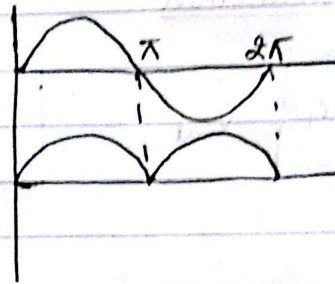
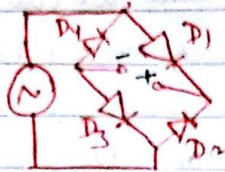


# LECTURE : 11

Date  
Page

## (b) Full wave Rectifier :

Observation

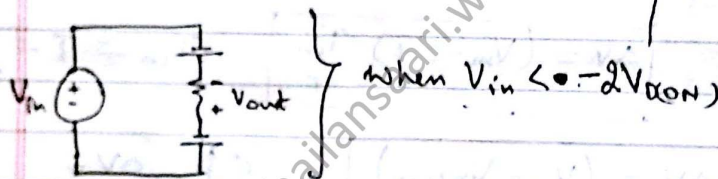
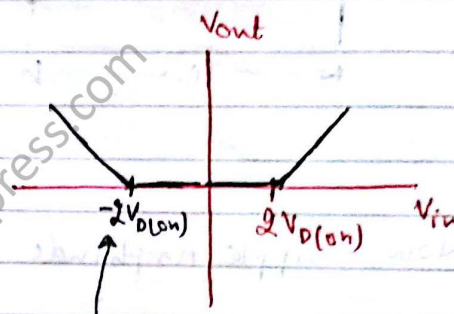


From 0 to  $\pi$   $D_1$  &  $D_3$  ON

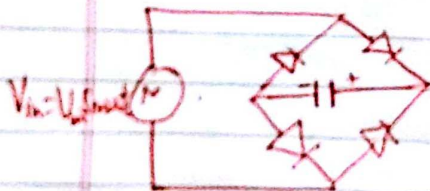
$$V_{out} = V_{in} - 2V_{D(on)}$$

From  $\pi$  to  $2\pi$ ,  $D_2$  &  $D_4$  ON

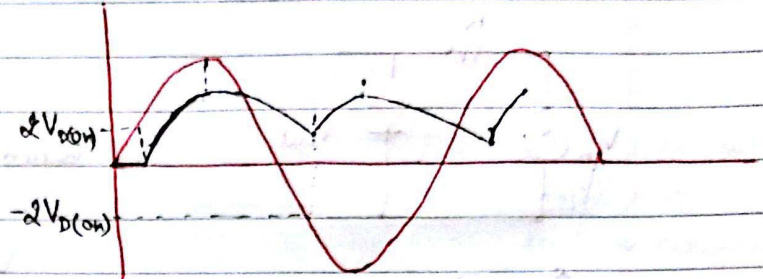
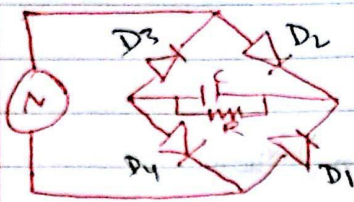
$$V_{out} = -V_{in} - 2V_{D(on)}$$



## FWR with Capacitor



When Voltage across both diode =  $2V_{D(on)}$  then Capacitor charges to max. value. After that diode gets off and the voltage across capacitor remains constant given by  $V_{out} = V_{in} - 2V_{D(on)}$

FWR with Capacitor & load

When  $V_{in} > 2V_{D(on)}$  Capacitor charges to max. value after that Diode turn off & Capacitor discharges till  $D_2$  &  $D_4$  are not ON. ~~then~~  $D_2$  &  $D_4$  ~~are~~ on when  $V_{in} = -2V_{D(on)}$  then again Capacitor charges.

Ripple in Amplitude

For HWR  $\rightarrow \Delta V = \frac{V_m - V_{D(on)}}{R_1 C_1} T_{in}$

For FWR  $\rightarrow \Delta V = \frac{V_m - 2V_{D(on)}}{R_1 C_1} \left( \frac{T_{in}}{2} \right)$

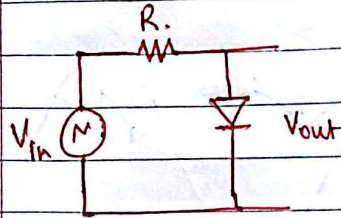
Formula from GATE point of view:

1.  $V_{avg} = V_o = \frac{2V_m}{\pi}$
2.  $V_{rms} = \frac{V_m}{\sqrt{2}}$
3.  $V_{PR} = V_m = \text{Peak inverse voltage of Diode}$
4.  $\Delta V = \frac{V_m}{2RCf}$  (for ideal diode)
5.  $f_{out} = 2f_{in}$



# LECTURE : 12.

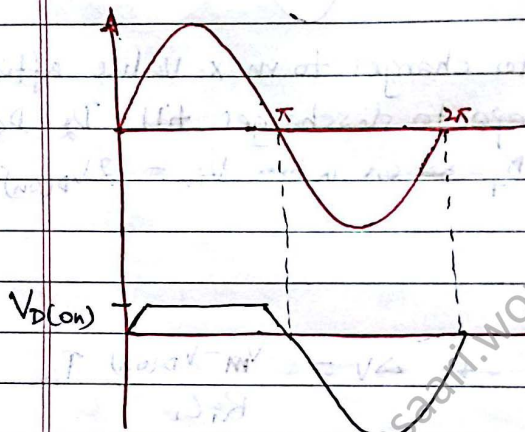
## LIMITING CIRCUITS



When  $V_{in} > V_{D(on)}$

Diode gets ON

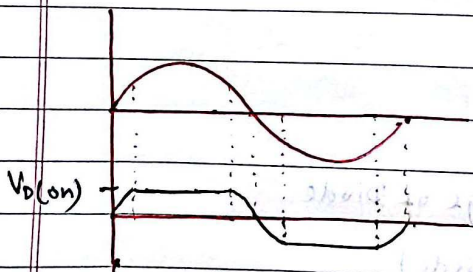
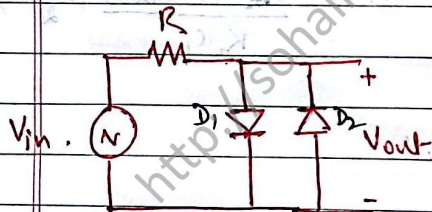
$$V_{out} = V_{D(on)}$$



When  $V_{in} < V_{D(on)}$

Diode  $\rightarrow$  O.C.

$$V_{out} = V_{in}$$



When  $V_{in} > V_{D1(on)}$

$D_1 \rightarrow ON$   $V_{out} = V_{D1(on)}$

When  $V_{in} < -V_{D2(on)}$

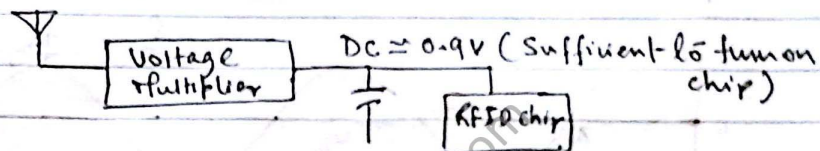
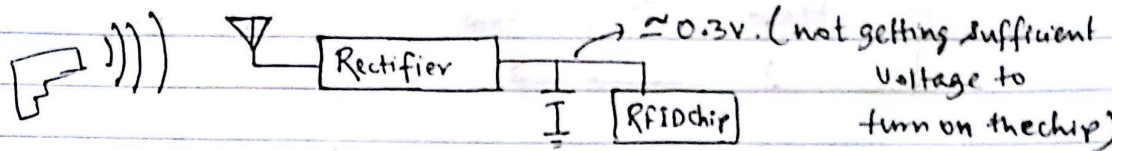
$D_2 \rightarrow ON$

$$V_{out} = -V_{D2(on)}$$

We limit the output voltage just by inserting diode in a proper manner.

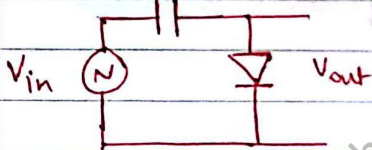
## VOLTAGE DOUBLER

Application:  
RFID

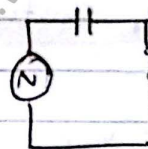


Voltage Doubler:

Example c

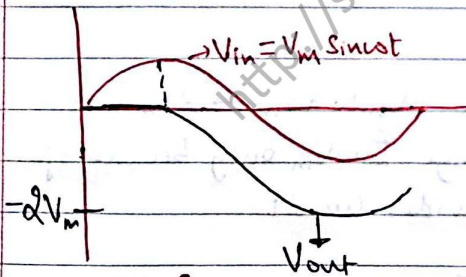


When  $V_{in} > 0$   $D \rightarrow$  short circuit  
Capacitor charges to max. value.  
 $V_c = V_m$

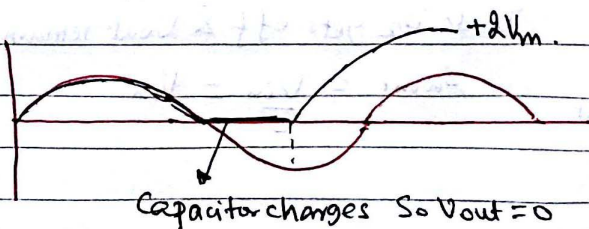
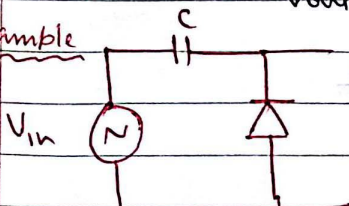


When Capacitor charges to  $V_m$   
&  $V_{in} \downarrow$  the Diode O.C.

$$V_{out} = -V_m - V_m \sin \omega t$$

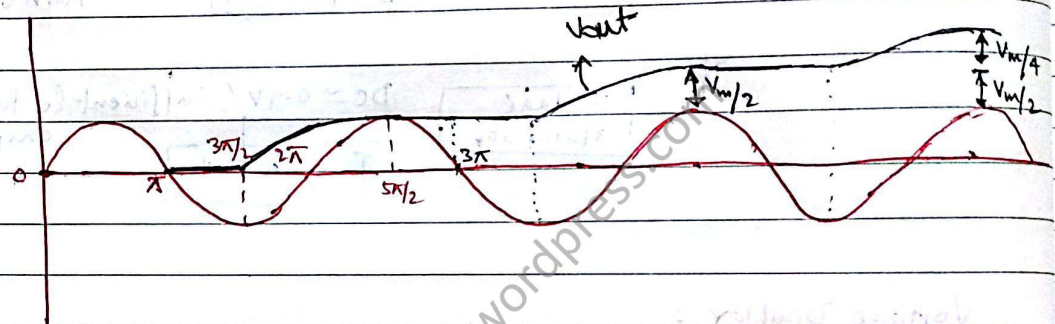
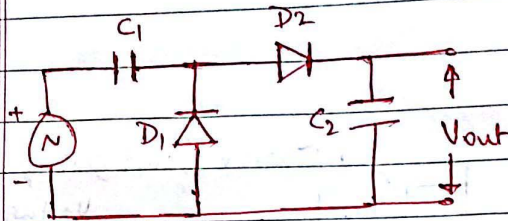


Example

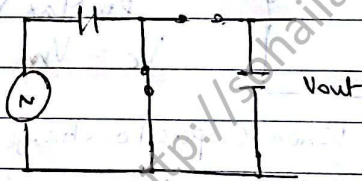




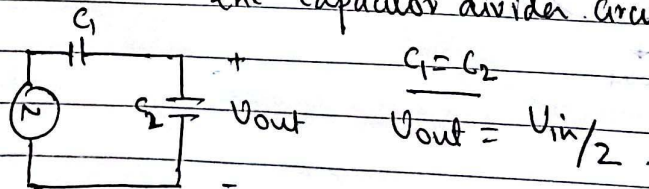
## Voltage Doubler Circuit



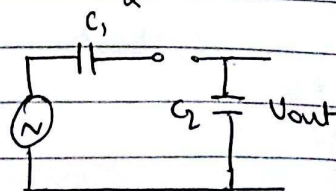
→ For simplicity let us consider circuit starts from  $\pi$ .  
From  $\pi - 3\pi/2$ ,  $D_1$  short circuit & capacitor charges to  $V_m$  value.



$\frac{3\pi}{2} \text{ to } \frac{5\pi}{2}$  input changes from  $-V_m$  to  $+V_m \rightarrow 2V_m$ .  
∴ Output will change to  $V_m$  only because of the capacitor divider circuit.



From  $\frac{5\pi}{2}$



Diode  $D_2$  gets off &  $V_{out}$  remains same =  $\frac{V_{in}}{2} = V_m$ .

The  $V_{out}$  remain equal to  $V_m$  at till  $7\pi/2$ . After that  $D_1$  turns ~~on~~ and  $D_1$  turns on at  $3\pi$  &  $C_1$  charges to  $V_m$ .

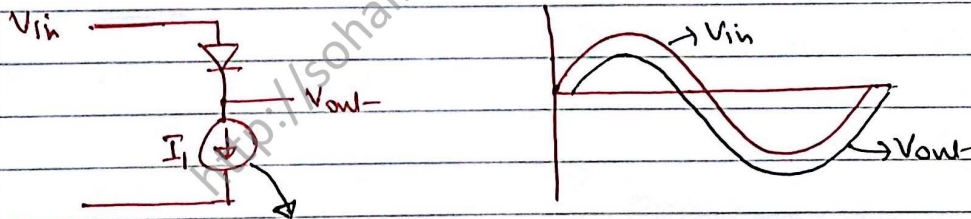
From  $7\pi/2$   $D_1$  gets off & the  $V_{out}$  increases to  $V_m/2$  and continues increasing as shown in waveform.  
 $\therefore$  The value of voltage across output

$$V_{out} = V_m + \frac{V_m}{2} + \frac{V_m}{4} + \frac{V_m}{8} + \dots$$

$$V_{out} = 2V_m$$

After some cycle  $V_{out} \approx 2V_m$  & becomes Constant.

### LEVEL SHIFT



Constant Current Source signifies that the diode is always ON.